

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 1: V–I Characteristics of PN Junction & Zener Diodes, and Study of Clipper & Clamper Circuits

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1. Aim

1. To plot the forward- and reverse-bias V – I characteristics of a PN junction diode and determine its cut-in voltage (V_γ), static forward resistance (R_{DC}), dynamic forward resistance (r_d), and reverse saturation current (I_s).
2. To plot the forward- and reverse-bias V – I characteristics of a Zener diode (1N4733A) and determine its Zener breakdown voltage (V_Z) and Zener resistance (R_Z).
3. To design, simulate, and test positive, negative, and biased (two-level) diode clipper circuits and observe their clipping levels.
4. To design, simulate, and test positive and negative diode clamper circuits and measure the resulting DC voltage shift.

2. Apparatus and Software Tools Required

Table 1: Equipment and Components Specifications

S.No.	Component / Equipment	Specification	Quantity
1	PN Junction Diode	1N4007 / 1N4148	1
2	Zener Diode	1N4733A ($V_Z = 5.1$ V, 1 W)	1
3	Resistors	1 k Ω , 10 k Ω , 100 k Ω	As required
4	Capacitor	0.1 μ F	1
5	DC Voltage Sources	0–30 V Variable, +1 V, –1 V Reference	As required
6	Function Generator / AC Source	1 kHz Sine wave, 4 V _{pp}	1
7	Simulation Software	LTspice	–
8	Digital Storage Oscilloscope (DSO)	Hardware measurement	1

3. Part A: PN Junction Diode Characteristics

3.1 Theoretical Background

A PN junction diode exhibits non-linear current-voltage characteristics described by the Shockley diode equation:

$$I = I_s \left(e^{\frac{V}{\eta V_T}} - 1 \right) \quad (1)$$

where I_s is the reverse saturation current, $V_T = \frac{kT}{q}$ is the thermal voltage (≈ 26 mV at room temperature), and η is the ideality factor.

- **Forward Bias:** Below cut-in voltage ($V_\gamma \approx 0.6\text{--}0.7$ V for Silicon), current is negligible. Above V_γ , current increases exponentially.
- **Reverse Bias:** Barrier height increases, allowing only minute reverse saturation current I_s to flow until reverse breakdown occurs.

3.2 Circuit Diagrams and Simulation Waveforms

3.2.1 PN Junction Diode – Forward Bias

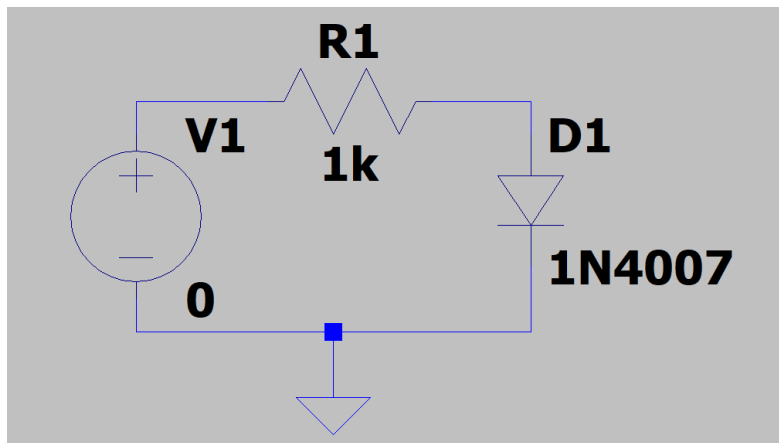
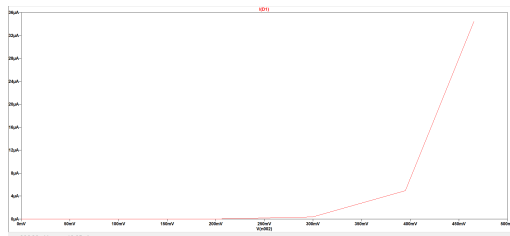
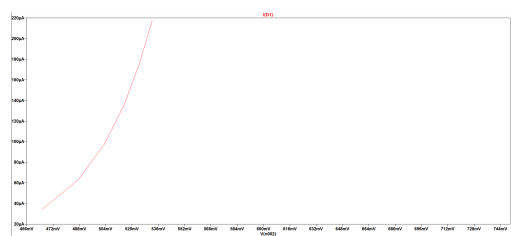


Figure 1: PN Junction Diode Forward Bias Circuit Schematic.



(a) Low Voltage Sweep (0 V to 0.5 V)



(b) Knee Region Sweep (0.5 V to 0.75 V)

Figure 2: PN Diode Forward V - I Characteristics Simulation Plots.

3.2.2 PN Junction Diode – Reverse Bias

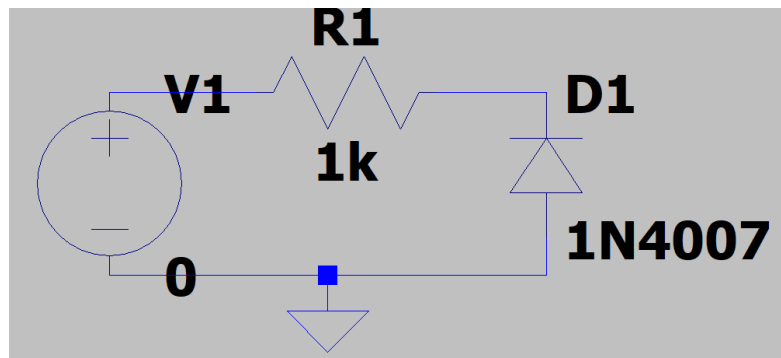


Figure 3: PN Junction Diode Reverse Bias Circuit Schematic.

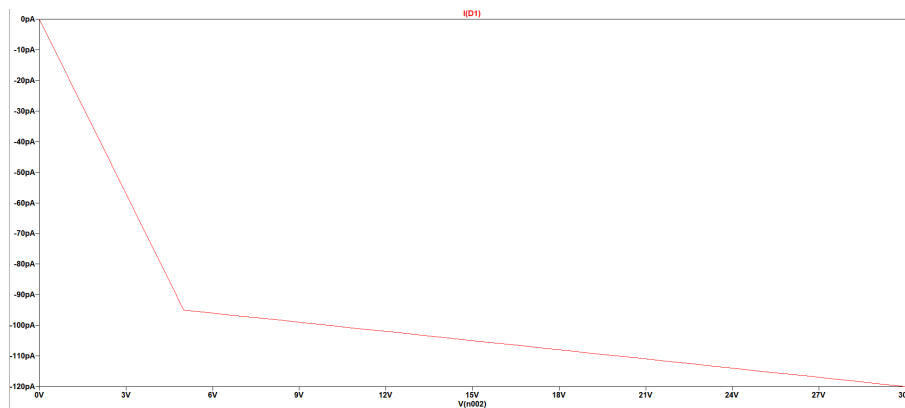


Figure 4: PN Diode Reverse Characteristic (0 V to 30 V).

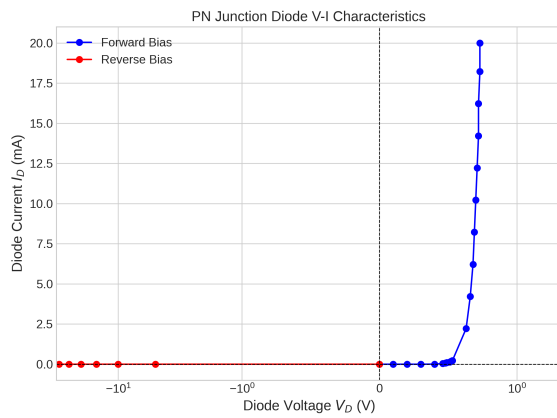
3.3 Experimental Observations (PN Diode)

Table 2: PN Junction Diode (1N4007) Forward and Reverse Bias Tabulated Readings

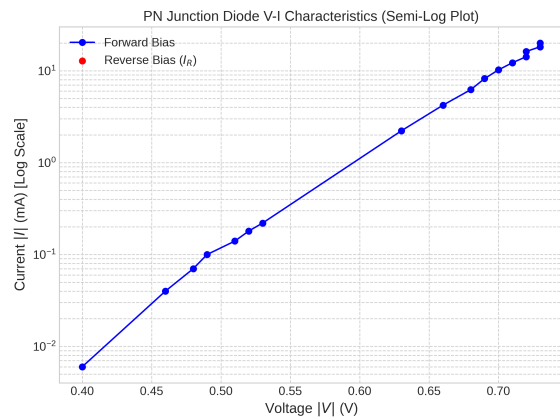
Forward Bias				Reverse Bias		
S.No.	V_{sup} (V)	V_f (V)	I_f (mA)	S.No.	$V_{sup} = V_r$ (V)	I_r (μ A)
1	0.00	0.00	0.000	1	0.0	0.0
2	0.40	0.40	0.006	2	-5.0	0.0
3	0.55	0.48	0.070	3	-10.0	0.0
4	0.65	0.51	0.140	4	-15.0	0.0
5	0.75	0.53	0.220	5	-20.0	0.0
6	2.88	0.63	2.220	6	-25.0	0.0
7	4.93	0.66	4.220	7	-30.0	0.0
8	6.97	0.68	6.220			
9	11.04	0.70	10.220			
10	15.11	0.72	14.220			
11	20.97	0.73	20.000			

3.4 Experimental V-I Characteristics & Plots

The experimental readings were plotted using Matplotlib in both linear scale (I vs V) and semi-log scale ($\log |I|$ vs V).



(a) Linear Scale Plot (I vs V)



(b) Semi-Log Plot ($\log |I|$ vs V)

Figure 5: Experimental V - I Characteristic Curves for PN Junction Diode (1N4007).

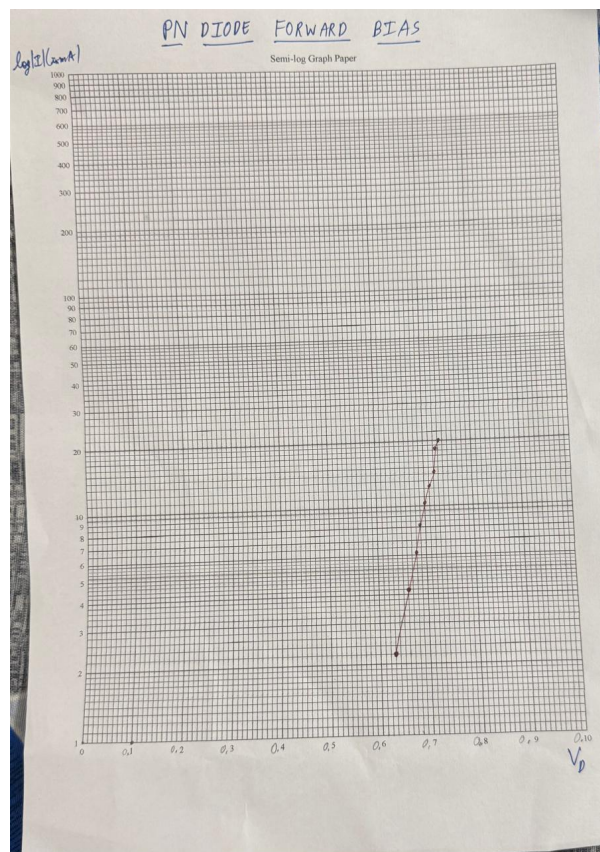


Figure 6: Graph-Paper Plot of the Forward-Bias V - I Characteristics of PN Junction Diode.

3.5 Calculated Parameters (PN Diode)

- **Cut-in Voltage (V_γ):** ≈ 0.60 V (exponential current rise begins above 0.6 V).
- **Static Forward Resistance ($R_{DC} = V_f/I_f$):**
 - At $I_f = 2.22$ mA, $V_f = 0.63$ V: $R_{DC} = \frac{0.63}{2.22 \text{ mA}} = 283.78 \text{ } \Omega$.
 - At $I_f = 4.22$ mA, $V_f = 0.66$ V: $R_{DC} = \frac{0.66}{4.22 \text{ mA}} = 156.40 \text{ } \Omega$.
 - At $I_f = 10.22$ mA, $V_f = 0.70$ V: $R_{DC} = \frac{0.70}{10.22 \text{ mA}} = 68.49 \text{ } \Omega$.
 - At $I_f = 20.0$ mA, $V_f = 0.73$ V: $R_{DC} = \frac{0.73}{20.0 \text{ mA}} = 36.50 \text{ } \Omega$.
- **Dynamic Forward Resistance ($r_d = \Delta V_f / \Delta I_f$):**
 - In knee region (2.22 mA to 4.22 mA): $r_d = \frac{0.66 - 0.63}{4.22 \text{ mA} - 2.22 \text{ mA}} = \frac{0.03 \text{ V}}{2 \text{ mA}} = 15.00 \text{ } \Omega$.
 - In high-conduction region (10.22 mA to 20.0 mA): $r_d = \frac{0.73 - 0.70}{20.0 \text{ mA} - 10.22 \text{ mA}} = \frac{0.03 \text{ V}}{9.78 \text{ mA}} = 3.07 \text{ } \Omega$.
- **Reverse Saturation Current (I_s):** $I_s < 1 \text{ } \mu\text{A}$ (0 mA measured from 0 V to 30 V).

4. Part B: Zener Diode Characteristics

4.1 Theoretical Background

A Zener diode is designed to operate safely in reverse breakdown (V_Z). Once in breakdown, the terminal voltage remains virtually constant over a wide range of reverse currents. In forward bias, it behaves as a normal PN junction diode.

4.2 Circuit Diagrams and Simulation Waveforms

4.2.1 Zener Diode – Forward Bias

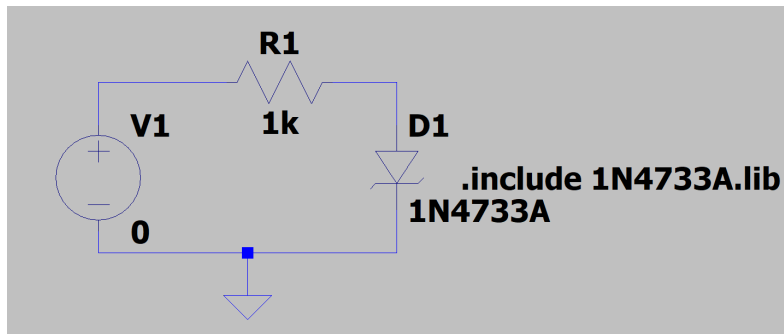


Figure 7: Zener Diode Forward Bias Circuit Schematic.

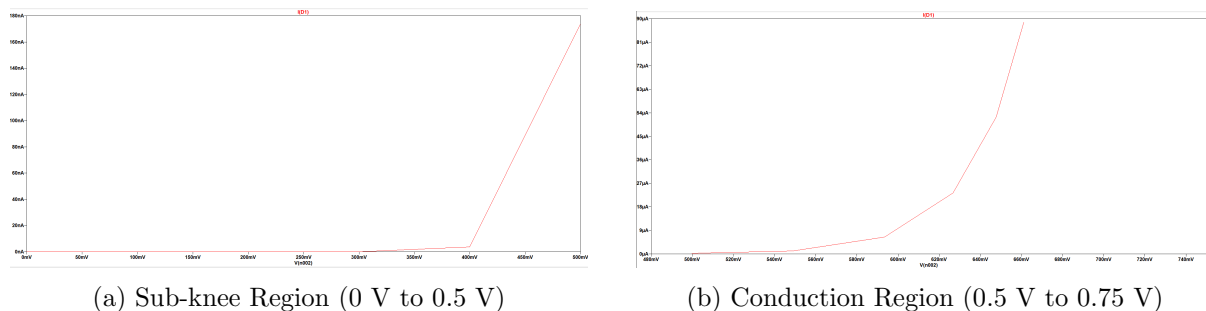


Figure 8: Zener Diode Forward Bias Simulation Plots.

4.2.2 Zener Diode – Reverse Bias and Breakdown

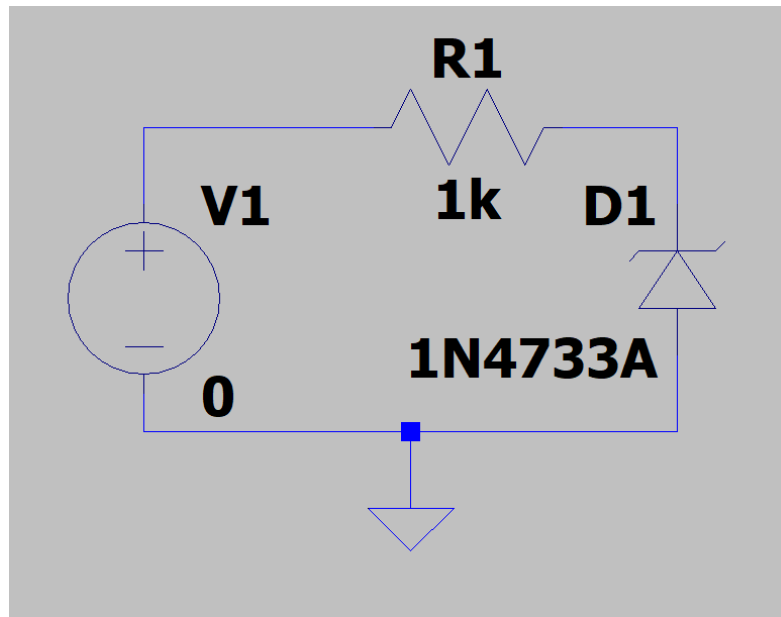


Figure 9: Zener Diode Reverse Bias Circuit Schematic.

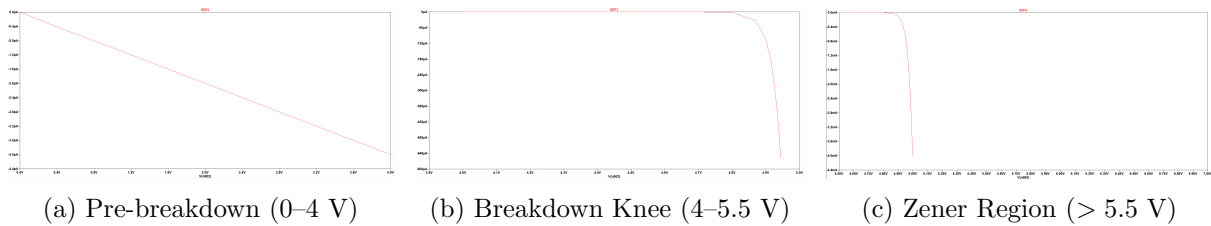


Figure 10: Zener Diode Reverse Breakdown Simulation Response.

4.3 Experimental Observations (Zener Diode)

Table 3: Zener Diode (1N4733A) Forward and Reverse Bias Tabulated Readings

Forward Bias				Reverse Breakdown Bias			
S.No.	V_{sup} (V)	V_f (V)	I_f (mA)	S.No.	V_{sup} (V)	V_r (V)	I_r (mA)
1	0.00	0.00	0.000	1	0.00	0.00	0.000
2	0.10	0.10	0.000	2	-1.00	-1.00	0.000
3	0.30	0.30	0.000	3	-2.50	-2.50	-0.001
4	0.50	0.50	0.000	4	-3.50	-3.48	-0.020
5	0.55	0.55	0.002	5	-4.10	-4.00	-0.100
6	0.60	0.59	0.008	6	-4.70	-4.40	-0.300
7	0.65	0.62	0.025	7	-5.10	-4.58	-0.520
8	0.70	0.65	0.050	8	-5.50	-4.70	-0.800
9	0.75	0.66	0.090	9	-14.04	-5.17	-8.800
				10	-25.43	-5.25	-20.000

4.4 Experimental V-I Characteristics & Plots

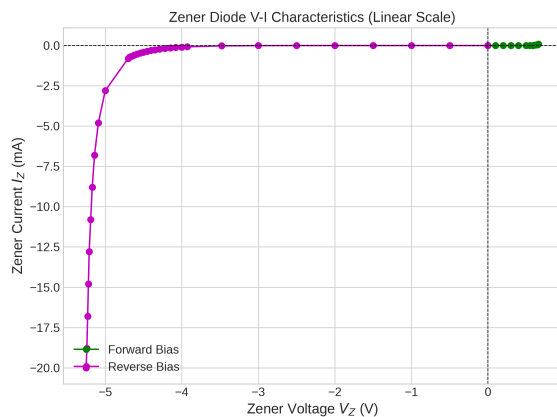
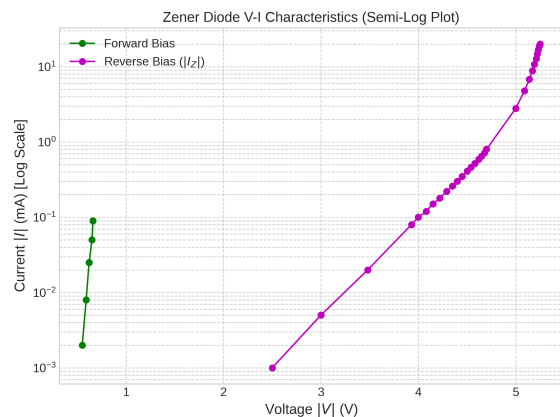
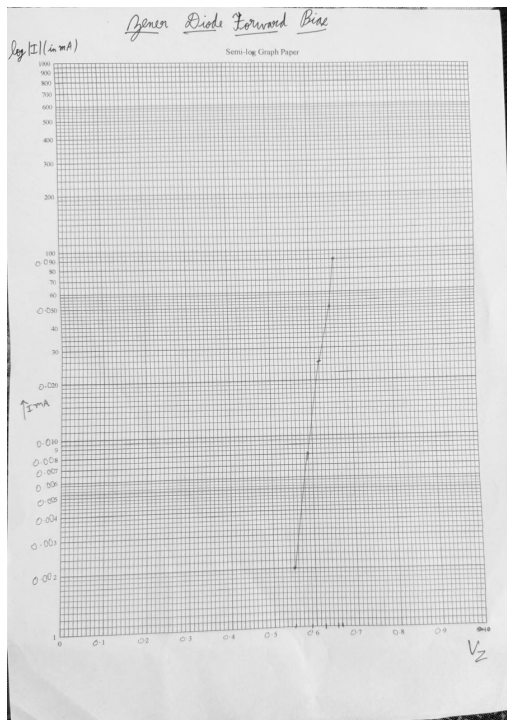
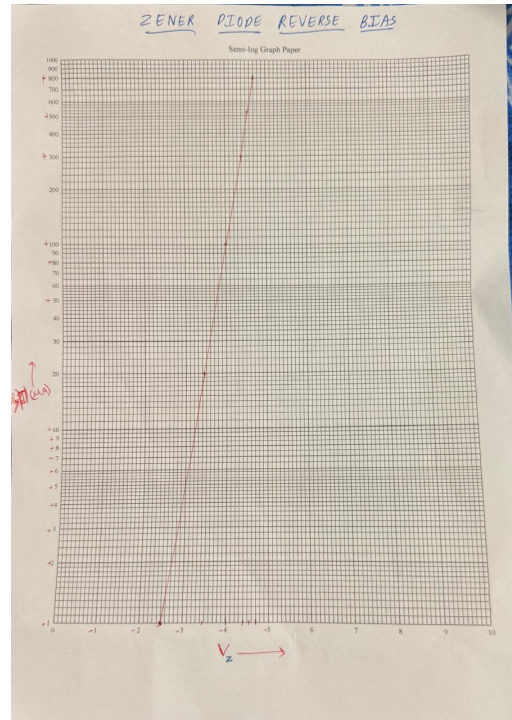
(a) Linear Scale Plot (I vs V)(b) Semi-Log Plot ($\log |I|$ vs V)Figure 11: Experimental V - I Characteristic Curves for Zener Diode (1N4733A).(a) Graph-Paper Plot of the Forward-Bias V - I Characteristics.(b) Graph-Paper Plot of the Reverse-Bias V - I Characteristics.

Figure 12: Graph-Paper Plots for Zener Diode (1N4733A).

4.5 Calculated Parameters (Zener Diode)

- **Zener Breakdown Voltage (V_Z):** Breakdown starts at $V_Z \approx 5.00$ V and stabilizes up to 5.25 V at 20 mA.
- **Zener Dynamic Resistance ($R_Z = \Delta V_Z / \Delta I_Z$):**

$$R_Z = \frac{5.25 \text{ V} - 5.00 \text{ V}}{20.0 \text{ mA} - 2.8 \text{ mA}} = \frac{0.25 \text{ V}}{17.2 \text{ mA}} = 14.53 \Omega \quad (2)$$

5. Part C: Diode Clipper Circuits

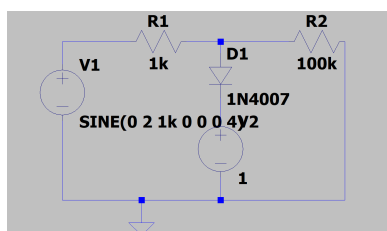
5.1 Theoretical Background

Clippers remove portions of an applied signal above or below chosen reference levels without distorting the remainder. A load resistor of $R_L = 100\text{ k}\Omega$ was connected across the output to ensure smooth DSO readings and eliminate loading effects.

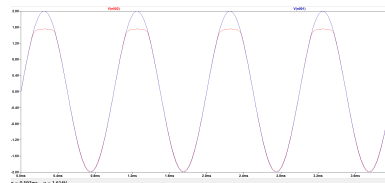
5.2 Circuit Diagrams, Simulations & Hardware DSO Waveforms

5.2.1 Positive Clipper (Clipping above +1 V reference)

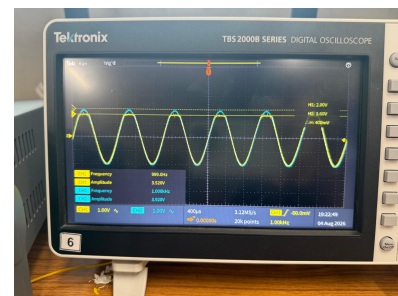
In positive clipping, the diode conducts when $V_{in} > +1\text{ V} + V_\gamma$, holding output to $+1.60\text{ V}$.



(a) Circuit Schematic



(b) LTspice Simulation

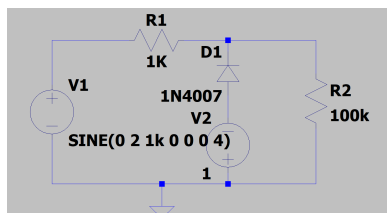


(c) DSO Trace

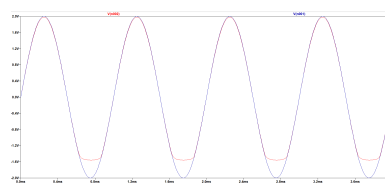
Figure 13: Positive Clipper Circuit, Simulation, and Measured DSO Waveform (Clipped at $+1.60\text{ V}$).

5.2.2 Negative Clipper (Clipping below -1 V reference)

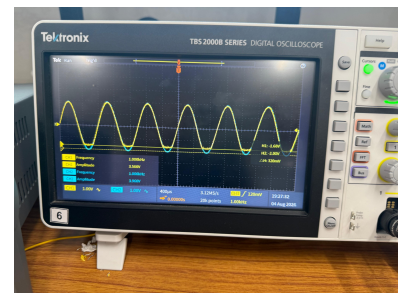
In negative clipping, the diode conducts when $V_{in} < -1\text{ V} - V_\gamma$, holding output to -1.68 V .



(a) Circuit Schematic



(b) LTspice Simulation

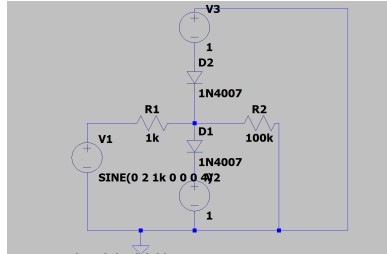


(c) DSO Trace

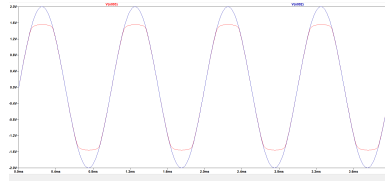
Figure 14: Negative Clipper Circuit, Simulation, and Measured DSO Waveform (Clipped at -1.68 V).

5.2.3 Biased Clipper (Two-Level Clipper)

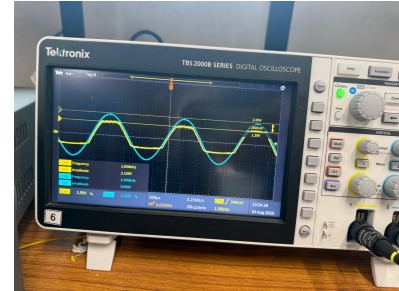
Both positive and negative peaks are clipped independently. The upper limit clipped at $+1.20\text{ V}$ and lower limit clipped at -0.92 V .



(a) Circuit Schematic



(b) LTspice Simulation



(c) DSO Trace

Figure 15: Biased Clipper Circuit, Simulation, and Measured DSO Waveform (Clipped at $+1.20\text{ V}$ & -0.92 V).

5.3 Clipper Experimental Observation Summary

Table 4: Diode Clipper Circuits Observation Table ($R_L = 100\text{ k}\Omega$)

Circuit	Input V_{pp} (V)	Theoretical Clip Level	Measured Clip Level	Inference
Positive Clipper	4.0	$+1.0\text{ V} + V_\gamma \approx +1.65\text{ V}$	Clip at $+1.60\text{ V}$	Upper peak clipping
Negative Clipper	4.0	$-1.0\text{ V} - V_\gamma \approx -1.65\text{ V}$	Clip at -1.68 V	Lower peak clipping
Biased Clipper	4.0	$+1.65\text{ V}$ & -1.65 V	$+1.20\text{ V}$ & -0.92 V	Dual-level clipping

6. Part D: Diode Clamper Circuits

6.1 Theoretical Background

Clampers shift an AC waveform vertically by clamping one peak to a fixed DC level without distorting the peak-to-peak amplitude. A load resistor $R_L = 100\text{ k}\Omega$ was connected across the output node to ensure smooth readings and proper discharge behavior of capacitor $C = 0.1\text{ }\mu\text{F}$.

6.2 Circuit Diagrams, Simulations & Hardware DSO Waveforms

6.2.1 Positive Clamper

The positive clamper shifts the waveform upward such that the negative peak is clamped near $-V_\gamma = -0.52\text{ V}$.

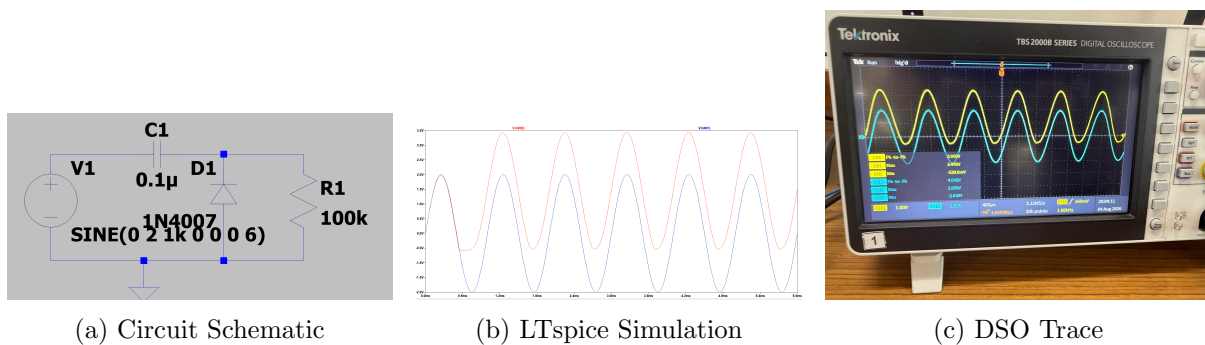


Figure 16: Positive Clamper Circuit, Simulation, and Measured DSO Waveform (Min level at -0.52 V).

6.2.2 Negative Clamper

The negative clamper shifts the waveform downward such that the positive peak is clamped near $+V_\gamma = +0.56\text{ V}$.

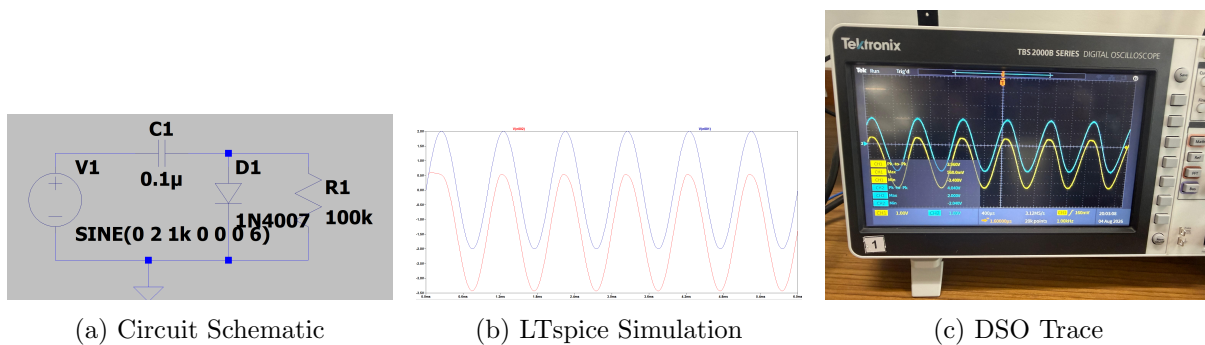


Figure 17: Negative Clamper Circuit, Simulation, and Measured DSO Waveform (Max level at $+0.56\text{ V}$).

6.3 Clamper Experimental Observation Summary

Table 5: Diode Clamper Circuits Observation Table ($R_L = 100 \text{ k}\Omega$)

Circuit	Input V_{pp} (V)	Theoretical Shift	Measured Clamped Level
Positive Clamper	4.0	Shift up by $+V_p - V_\gamma \approx +1.30 \text{ V}$	Min level at -0.52 V
Negative Clamper	4.0	Shift down by $-V_p + V_\gamma \approx -1.30 \text{ V}$	Max level at $+0.56 \text{ V}$

7. Discussion & Overall Inferences

1. **PN Diode Non-linearity & Conduction:** Forward bias readings demonstrate exponential growth after overcoming cut-in barrier voltage $V_\gamma \approx 0.60 \text{ V}$. Semi-log plots confirm Shockley's relation with linear slope in forward conduction. Static resistance decreases from $R_{DC} = 283.78 \Omega$ at 2.22 mA to $R_{DC} = 36.50 \Omega$ at 20.0 mA , while dynamic resistance settles at $r_d = 3.07 \Omega$.
2. **Zener Breakdown Regulation:** In reverse bias, breakdown occurs sharply at $V_Z \approx 5.00\text{--}5.25 \text{ V}$. The low dynamic Zener resistance $R_Z = 14.53 \Omega$ confirms excellent voltage regulation stability under varying reverse currents.
3. **Clipper Behavior:** Incorporating a $100 \text{ k}\Omega$ load resistor prevented signal attenuation. Clipping thresholds closely match $V_{ref} \pm V_\gamma$ ($+1.60 \text{ V}$ for positive, -1.68 V for negative, and $+1.20 \text{ V} / -0.92 \text{ V}$ for biased clipper).
4. **Clamper DC Restoration:** Positive and negative clampers successfully restore the DC level, shifting the minimum peak to -0.52 V and maximum peak to $+0.56 \text{ V}$ respectively without altering the peak-to-peak amplitude.